

CSA1013-software engineering

Name :M.hanif Riyaz

Reg no:192521123

Faculty name:DR.R.latha

Smart Airport Passenger Flow and Queue Management System

Problem Statement

Airports worldwide experience heavy passenger traffic, especially during peak hours, leading to congestion at check-in counters, security checkpoints, immigration, and boarding gates. Inefficient queue management results in long waiting times, passenger dissatisfaction, and flight delays. Existing systems rely heavily on manual monitoring, which lacks real-time insights and predictive capabilities. The proposed Smart Airport Passenger Flow and Queue Management System leverages Artificial Intelligence (AI), Computer Vision, Internet of Things (IoT), and cloud computing to monitor passenger density, predict congestion, and optimize queue management. The system aims to enhance operational efficiency while improving passenger experience.

Objectives of the Proposed System

The primary objective of the system is to improve airport operations by intelligently managing passenger flow using advanced technologies. The system continuously monitors passenger movement in real time using cameras and IoT sensors. It predicts congestion using AI-based models and dynamically allocates airport resources such as staff and counters. The platform generates real-time alerts, analyzes passenger flow patterns, enhances security checkpoint efficiency, and ensures smooth passenger movement throughout the airport.

Scope of the Problem

The system is designed for airports, airline operators, security agencies, and airport management authorities. It provides real-time monitoring of passenger density, queue lengths, and waiting times across different airport zones. The platform can be

implemented in domestic and international airports, smart airports, and large transportation hubs. It also supports future integration with biometric systems, smart boarding systems, and smart city infrastructure to improve overall transportation efficiency.

Stakeholders, Roles, and Responsibilities

Airport authorities are responsible for managing airport operations and ensuring smooth passenger flow. Airline staff handle check-in, boarding, and customer service. Security personnel monitor checkpoints and ensure passenger safety. System administrators maintain software, databases, and cloud infrastructure. Passengers benefit from reduced waiting times and improved travel experience. Data analysts and researchers use system data to improve predictive models and operational strategies.

Challenges Faced by Stakeholders

Airports face challenges such as unpredictable passenger flow, peak-hour congestion, and inefficient resource allocation. Manual monitoring systems lack real-time insights and predictive capabilities. Security checkpoints often become bottlenecks due to high passenger density. Infrastructure limitations and high operational costs further complicate queue management. Additionally, integrating advanced technologies with existing airport systems can be complex and expensive.

3. Analyze the Current Approach

Existing System

Current airport queue management systems rely on manual observation, CCTV monitoring, and basic scheduling techniques. Passenger flow is managed based on staff experience and predefined procedures. Although some airports use digital display systems, they lack

predictive capabilities and real-time optimization.

Strengths and Limitations

The existing system is simple and cost-effective but lacks automation and intelligence. It does not provide real-time congestion analysis or predictive insights. Manual decision-making leads to inefficiencies, increased waiting times, and poor passenger experience.

Problems and Gaps

Major gaps include the absence of AI-based congestion prediction, lack of real-time monitoring, inefficient resource allocation, delayed response to congestion, and limited data analytics capabilities. These issues reduce operational efficiency and increase passenger dissatisfaction.

4. Proposed Software-Based Solution

The proposed solution is an AI-powered Smart

Airport Passenger Flow and Queue Management System. It integrates Computer Vision, IoT sensors, and cloud analytics to monitor passenger density in real time. AI algorithms analyze data to predict congestion and recommend optimal resource allocation. The system provides dashboards, alerts, and reports for airport authorities, enabling data-driven decision-making and efficient operations.

Major Functional Modules

- User Management Module – Handles authentication and role-based access.
- Passenger Monitoring Module – Tracks passenger movement using cameras and sensors.
- Congestion Prediction Module – Uses AI algorithms to predict crowd density.
- Queue Optimization Module – Suggests optimal queue management strategies.

- Resource Allocation Module – Dynamically assigns staff and counters.
- Alert and Notification Module – Sends real-time congestion alerts.
- Reporting and Analytics Module – Generates reports and insights.
- Dashboard Module – Provides visual representation of passenger flow data.

Functional and Non-Functional Requirements

Functional requirements include real-time monitoring, congestion prediction, queue optimization, alert generation, and reporting.

Non-functional requirements include scalability, reliability, security, fast response time, high availability, and user-friendly interface.

Application of Software Engineering Principles

The system follows modularity by dividing into independent modules such as monitoring,

prediction, and reporting. Scalability ensures support for large passenger volumes. Maintainability allows easy updates and improvements. Reliability ensures accurate predictions and continuous operation. Security is maintained through authentication, encryption, and secure cloud storage.

5. Justification of Methodology

The Agile Software Development Life Cycle (SDLC) is suitable for this system due to its flexibility and iterative development approach. Continuous feedback from stakeholders allows improvements in features and performance. Agile ensures faster development, better quality, and easy integration of new technologies.

6. Expected Outcomes and Limitations

Expected Outcomes

The system is expected to reduce passenger

waiting time, improve airport efficiency, optimize resource utilization, enhance passenger experience, reduce flight delays, enable real-time monitoring, support data-driven decisions, and improve overall service quality.

Limitations

The system requires high initial investment for IoT devices and AI infrastructure. Accuracy depends on data quality and sensor performance. Network failures and cybersecurity risks may affect reliability. Integration with existing airport systems may require technical expertise.